Gennaio 2026

1. Public Key Infrastructure and Digital Signatures

Which of the following sentences are true? (Select all statements that are true.)

- [x] A. A digital certificate cryptographically binds a public key to the identity of its owner. → certification authority firma digitalmente un documento che contiene chiave pubblica + dati ident.
- [ ] B. The certification authority stores users' private keys in order to verify digital signatures. → la verifica avviene con chiave pubblica, privata deve rimanere segreta
- [x] C. A smartcard can generate a digital signature without revealing the private key. → Si, PC invia a smartcard l'impronta (hash) del documento e smartcard esegue calcolo matematico con chiave priv.
- [ ] D. Digital signatures can only be verified using specialized hardware devices. → tramite Software
- [ ] E. Digital signatures must be stored on the smartcard that generated them. → firma digitale stringa di bit allegata al documento firmato

2. Public Key Cryptography – Diffie-Hellman Please justify your answers.

(a) Which of the following choices of q and α are appropriate? (For the security of the scheme, q and α must be considerably larger; however, you can neglect this aspect of the problem.)

- [ ] A. $q = 6, \alpha = 3$
- [ ] B. $q = 6, \alpha = 5$
- [ ] C. $q = 7, \alpha = 2$
- [x] D. $q = 7, \alpha = 5$

You may find it useful to use the following (partial tabulation) of the modular exponentiation function.

x	$3^x \bmod 6$	$5^x \bmod 6$	$2^x \bmod 7$	$5^x \bmod 7$
1	3	5	2	5
2	3	1	4	4
3	3	5	1	6
4	3	1	2	2
5	3	5	4	3
6	3	1	1	1

(b) Assume the values of q and α identified in the previous question. If $Y_A = 4$ is the public key of A, what is the private key X_A of A?

$Y_A = \alpha^{X_A} \bmod q$
$4 = 5^{X_A} \bmod 7$
$X_A = 2$

$5^2 \bmod 7 = 25 \bmod 5 = 4$

(c) If $Y_B = 5$ is the public key of B, what is the shared key K?

$K = Y_B^{X_A} \bmod q$
$K = 5^2 \bmod 7$
$K = 4$

3. Security Protocols Consider the following authentication protocol

1. $A \to B : E(K, [A, B, T])$

where K is a key shared between A and B and T is a time-stamp generated by A.

(a) Indicate which of the following security properties are met by the protocol. For the properties that are met by the protocol describe the assumptions that must hold for the property to hold. For the properties that are not met by the protocol write an execution trace witnessing the violation of the property.

- [ ] A. A authenticates B
- [x] B. B authenticates A
- [ ] C. Secrecy of N
- [x] D. Secrecy of K

(b) Describe the actions carried out by B upon receipt of the message for a proper functioning of the protocol.

B usa la chiave condivisa K per decriptare il messaggio e leggerne il contenuto. Verifica che il primo identificativo indichi il mittente, A, e che il secondo identificativo indichi che lui sia il destinatario. Controlla poi che il Timestamp T sia abbastanza recente.

(c) What are the consequences of using an implementation of the protocol where the timestamp T is very coarse, e.g. successive time instants correspond to large intervals (say several seconds).

Se un attaccante intercettasse il messaggio, potrebbe riutilizzarlo per eseguire un replay attack.

(d) Assume that successive time instants correspond to 1 second and that timestamps are represented by 12 bits. How long must an attacker wait to mount a replay attack?

- [ ] A. ~ 1 second
- [ ] B. ~ 1 minute
- [x] C. ~ 1 hour
- [ ] D. ~ 1 day
- [ ] E. ~ 1 week
- [ ] F. ~ 1 month

2^{12}

4. Mandatory Access Control

Consider the following security labeling in the Bell–LaPadula model:

Subject	Label
Alice	Confidential
Bob	Top Secret
Carol	Public
Dave	Secret

Object	Label
P1	Confidential
P2	Top Secret
P3	Public
P4	Secret
P5	Restricted

The security levels follow the ordering:

Top Secret > Secret > Restricted > Confidential > Public.

(a) Interpret the labels as security labels in the simplified Bell–LaPadula model. Complete the following **access control matrix** by writing, for each cell, what access Bell–LaPadula allows (`r`, `w`, `rw`, or `–`).

C TS P S R

Subject \ Object	P1	P2	P3	P4	P5
Alice (Confidential)	rw	w	r	w	w
Bob (Top Secret)	r	rw	r	r	r
Carol (Public)	w	w	rw	w	w
Dave (Secret)	r	w	r	rw	r